Rajiv Ranganath

rajivranganath3@gmail.com +91 94816 79510 Bangalore, India linkedin.com/in/rajiv-ranganath-8b0bb1285
github.com/RajivRanganath rajivranganath.github.io

Career Overview

Final-year Computer Science student (CGPA 9.36) building and evaluating LLM/RAG systems and real-time probabilistic estimation pipelines, including ensemble Kalman filtering and multi-target data association. Currently working at IIT Kharagpur on LLM-based ESG disclosure extraction and auditing evaluation against XBRL filings, with an emphasis on leakage-resistant, source-verified evaluation.

Education

B.N.M. Institute of Technology 2023 – 05/2027
B.E. Computer Science & Engineering Bangalore, Karnataka
CGPA: 9.36

Sri Kumarans PU College 2021 – 2023
Class XII (PCM), State Board · 92.3% Bangalore, Karnataka

Experience

AI Intern · IIT Kharagpur · Jul 2026 to Present · Remote
- Audited the inherited evaluation code, removed **fabricated data and answer-key leakage**, and rebuilt scoring so every answer must cite its source, with Camelot table checks and a 6-type hallucination classification.
- Built an LLM/RAG pipeline that extracts figures from company sustainability reports (BRSR) and certifies a value only when its source checks out; certified values match XBRL filings **97.5% (dev) and 93.3 to 100% on three of four locked holdouts**.
- Designed automatic fallback across five LLM providers (Groq → Gemini → NVIDIA NIM → OmniRoute → Ollama) with run-time model discovery, guarded by 337 pytest tests and GitHub Actions CI.
- Modelled whether carbon-credit certificates should expire under India's CCTS, using a **waste-heat-recovery investment model anchored to five notified cement plants** and benchmarked against linear and mixed-integer programs; five-agent PPO learners never reached a stable outcome, which is reported rather than hidden.

Deep Learning Intern · E2E Management Services · Jun 2025 · Bangalore
- 4-week applied deep learning and reinforcement learning programme; capstone: ResNet-152 diabetic retinopathy classifier.

Projects

Orbit Guard AI: Real-Time Space Debris Tracking & Collision Warning System
Tracks space debris from sparse ground-station observations and warns of likely collisions.
- **200-particle Ensemble Kalman Filter** that estimates orbits from noisy observations, modelling Earth's oblateness (J2), atmospheric drag and solar radiation pressure, with the integrator verified against two-body physics (~10^{-12} energy and momentum drift over 24 h).
- Built two observation-to-object matchers, Hungarian nearest-neighbour (GNN) for clean data and JPDA for clutter, plus collision alerts (B-plane), where JPDA beat GNN by **11–14 points** of association accuracy in synthetic 8-object clutter at 100% track stability.
- Stack: Python · NumPy · FastAPI · WebSocket · React · Three.js · Live demo

Cooperative MARL for Energy-Harvesting Sensor Scheduling
- Trained three cooperative RL methods (IQL, VDN, QMIX; 25,474-parameter GRU policies) against seven hand-written heuristics to schedule four solar-powered sensor nodes, using eight sanity checks, single-use 30-seed test splits and win criteria fixed in advance.
- Unanimous-vote QMIX ensemble gained +3.70 reward (95% CI [2.10, 5.30]), lost to a one-line threshold rule by 4.95 but showed better data freshness, while analytical sizing shows ONNX exports fit **15 of 17 microcontrollers** at 0.25–6.8 ms per decision.
- Stack: Python · PyTorch · EPyMARL · ONNX · pytest (67 tests)

Diabetic Retinopathy Classifier
- Trained a 5-class diabetic retinopathy grading model on the APTOS 2019 fundus image dataset using ResNet-152 with a custom 512-dimensional classification head.
- Stack: Python · PyTorch · ResNet-152 · APTOS 2019

Dual Engine Plagiarism Detector
- Built and deployed a two-stage detector: lexical scoring (TF-IDF + Jaccard) for exact-match detection against local documents, and semantic similarity via all-MiniLM-L6-v2 embeddings against live web sources; outputs sentence-level attribution with similarity scores and includes a paraphrase-rewriting module.
- Stack: Python · Scikit-learn · Sentence Transformers · all-MiniLM-L6-v2

Skills

- Languages: Python, SQL, C, Java
- Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib; state estimation (Kalman / ensemble filters), multi-target tracking
- Core CS (coursework): Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- Generative AI & LLM Systems: RAG architecture, LLM evaluation & benchmarking, XBRL / Camelot table extraction, FAISS, LangChain, Groq API, HuggingFace Transformers, Sentence Transformers
- Backend & Infrastructure: FastAPI, WebSocket, Docker, Git, MongoDB, React / Three.js
- RL & Testing: PPO, QMIX, ONNX, pytest, GitHub Actions

Certifications

- Introduction to Internet of Things, NPTEL (IIT Kharagpur) · Elite + Gold, top 5% · 93%